

PQC in QUIC and HTTP/3: Securing the Next-Generation Internet

Day 27 | Category: Network Protocol Integration

Date: 2026-09-20 |

DAILY MONITORING

PQC in QUIC and HTTP/3: Securing the Next-Generation Internet

As the internet transitions to faster and more efficient protocols like QUIC and HTTP/3, ensuring their cryptographic foundations are secure against future quantum threats is paramount. The integration of Post-Quantum Cryptography (PQC) into these modern protocols presents unique challenges and opportunities, primarily centered around maintaining low latency and high performance.

The Challenge: Latency in Connection Establishment

QUIC, which runs over UDP, is designed to reduce connection establishment latency compared to traditional TCP+TLS. A standard TLS 1.3 handshake requires one round-trip time (1-RTT). QUIC improves this with a 0-RTT connection resumption feature. The introduction of larger PQC key encapsulation mechanisms (KEMs) and digital signatures can increase the size of handshake packets, potentially negating QUIC's low-latency benefits.

Key Consideration: The larger ciphertext and public key sizes of ML-KEM, and especially the large signature sizes of SLH-DSA, can increase the amount of data exchanged during the initial handshake, which could lead to packet fragmentation and retransmissions, thereby increasing latency.

Hybrid Key Exchange: The Pragmatic Approach

Similar to TLS 1.3, the initial strategy for integrating PQC into QUIC is the hybrid key exchange model. This approach combines a classical key exchange algorithm (like X25519) with a post-quantum one (like ML-KEM-768). This ensures that the connection remains secure against classical attacks, while also providing protection against quantum adversaries.

- **Benefit 1 (Security):** It provides a safety net. If a flaw is discovered in the PQC algorithm, the connection is still protected by the well-vetted classical cryptography.
- **Benefit 2 (Compliance):** It allows for a gradual transition, meeting current security requirements while preparing for future ones.

Performance Metrics: PQC's Impact on QUIC

Recent research and experiments from organizations like Cloudflare and Google have focused on quantifying the impact of PQC on QUIC's performance. The table below summarizes the expected overhead.

Key Exchange Mechanism	Added Handshake Latency (ms)	Increase in Handshake Size (Bytes)	Primary Use Case
X25519 (Classical)	Baseline	Baseline (~100 B)	Standard TLS/QUIC
X25519 + ML-KEM-768 (Hybrid)	+5-15 ms	~1,200 B	Balanced Security and Performance
X25519 + ML-KEM-1024 (Hybrid)	+10-25 ms	~1,600 B	Higher Security, Moderate Performance Cost

Note: Latency figures are estimates and can vary significantly based on network conditions and server proximity.

The Road Ahead: Signatures and Authentication

While key exchange is the primary focus, server authentication using PQC digital signatures (ML-DSA, SLH-DSA) is the next frontier. The larger size of PQC signatures, particularly stateless hash-based ones like SLH-DSA, poses a more significant challenge for QUIC's performance goals. Research is ongoing to optimize certificate chains and signature aggregation to minimize this impact.

Daily Quiz

1. What is the primary challenge when integrating PQC into QUIC and HTTP/3?
2. Why is a hybrid key exchange model considered the most pragmatic approach for the initial PQC integration in QUIC?
3. According to the comparison table, what is the estimated increase in handshake size when using a hybrid approach with X25519 and ML-KEM-768?